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Original Articles

Seasonal variation of cultural ecosystem services in urban blue-green spaces based on the SoLVES model and social media data

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ABSTRACT

Cultural Ecosystem Services (CES) are an essential component of ecosystem services, playing a key role in enhancing human well-being. However, conventional CES perception studies are often restricted by static frameworks, limiting their ability to capture spatio-temporal dynamics. Although social media data provide advantages for tracking CES dynamics, platform-imposed spatial boundaries hinder fine-scale analysis. To address these limitations, this study introduces an innovative approach that extracts geolocation information from social media photographs to overcome platform constraints. A dynamic CES assessment framework was developed based on the Social Values for Ecosystem Services (SoLVES) model and applied to the North Section of Shanghai Botanical Garden to examine seasonal variations in public CES perception. The findings reveal: (1) CES perception exhibits distinct seasonal preferences. Near-water areas record higher CES values in summer, while vegetated areas far from roads decline notably in winter. Scenic sites with spring and summer blooms show significantly higher CES values in those seasons than in autumn and winter. (2) Core landscape features, such as the Submerged Trestle, consistently maintain high CES values year-round. (3) Road accessibility plays a decisive role in shaping CES perception. (4) Annual CES assessments are strongly dominated by spring data, which may obscure seasonal spatial patterns. This study advances the understanding of seasonal mechanisms influencing CES in plant-themed urban blue-green spaces and provides a scientific basis for adaptive design and management strategies.

1. Introduction

Urban blue-green spaces refer to the combination of urban green spaces and blue areas, serving multiple functions including ecological regulation, landscape beautification, and recreational activities (Duan et al., 2024; Yang et al., 2024; Zhou et al., 2024). They play a critical role in improving the urban ecological environment by reducing noise, regulating climate, mitigating the urban heat island effect, and increasing air humidity (Jaung et al., 2020; Quan et al., 2023; Zhang et al., 2025a,b). Furthermore, urban blue-green spaces provide residents with places to connect with nature (Wood et al., 2025), helping to alleviate stress, improve mental health, and enhance subjective well-being (Liu et al., 2022; Zhou et al., 2022). Cultural Ecosystem Services (CES), a vital component of ecosystem services, were proposed by the Millennium Ecosystem Assessment (MEA, 2005). They refer to the non-material benefits humans obtain from ecosystems, encompassing spiritual fulfillment, aesthetic experiences, cognitive development, recreation and relaxation, and place identity (Milcu et al., 2013). CES cover multiple dimensions such as aesthetic value, cultural heritage, social relations, and sense of place. They represent a crucial link between natural systems and human society (Jupe et al., 2025a). A comprehensive understanding of CES characteristics is essential for holistically promoting high-quality development. In the context of rapid urbanization, the role of CES in sustaining human well-being is particularly prominent (Xia et al., 2024). However, the provision of CES is significantly influenced by natural phenological rhythms. Their use and

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perception also exhibit distinct dynamic characteristics across spatial and temporal dimensions due to variations in human activities (Graves et al., 2017; Aiba et al., 2019; Graves et al., 2019; Vierikko and Yli-Pelkonen, 2019; Kulczyk et al., 2024). For instance, biological phenological processes (e.g., flowering periods, bird migration) and landscape attributes (e.g., vegetation color, water body conditions) change with seasons, thereby affecting the actual provision of CES. Concurrently, climatic conditions in different seasons influence the frequency and types of human outdoor activities (Elliott et al., 2019). Mild weather in spring and summer favors recreational activities, while low temperatures, precipitation, or snow cover in autumn and winter may suppress activity frequency. However, these same conditions may also stimulate seasonally specific demands, such as snow appreciation. Consequently, relying solely on survey data from a single static point in time makes it difficult to comprehensively reflect the seasonal evolution patterns of CES. As a result, their dynamic characteristics across the temporal dimension may be underestimated or overlooked.

CES possess characteristics such as intangibility, subjectivity, and lack of clearly defined categorical boundaries. Furthermore, their association with economics, sociology, and numerous other disciplines increases the difficulty of their valuation and identification (Tian et al., 2023; Adhikary et al., 2025). Regarding the perception of CES, traditional research has predominantly relied on methods such as questionnaire surveys, interviews, and field observations to gather data on public preferences (Yang and Cao, 2022). While these approaches offer advantages in capturing subjective perceptions, they suffer from several limitations, including high data acquisition costs, limited sample sizes, difficulties in covering large spatial extents, and susceptibility to influences from respondents' memory and expressive abilities. To address these shortcomings, the United States Geological Survey (USGS) developed the Social Values for Ecosystem Services (SoLVES) model as a key tool for the spatial quantification of CES. Based on a GIS platform, SoLVES transforms public evaluations of non-monetized CESs (such as aesthetics and recreation) into spatial representations, providing an effective pathway for the quantitative modeling of CES (Sherrouse and Semmens, 2020; Sherrouse et al., 2022). In recent years, the SoLVES model has been widely applied in research focused on CES assessment (Guan et al., 2023; Tian et al., 2024). However, the data utilized by the SoLVES model typically relies on static questionnaire surveys, making it difficult to reflect seasonal variations in public perception of CES. This limitation restricts the model's applicability for dynamic analysis at the seasonal scale. Although recent pioneering work, such as that by Huang et al. (2024), has begun to explore these seasonal dynamics, such studies remain scarce.

The rise of social media platforms has led to a vast number of users routinely documenting and sharing their recreational experiences and emotional feelings in urban green spaces (Li et al., 2023), providing a novel data source for CES research. Compared to traditional methods, social media data offer advantages such as low collection costs, long temporal coverage, and large data volumes, demonstrating particular potential for capturing year-round temporal variations. By integrating technologies such as natural language processing (NLP), keyword extraction, and sentiment analysis, it becomes possible to achieve dynamic quantification and spatiotemporal analysis of public CES perception. Numerous studies have already begun leveraging social media data to identify public aesthetic preferences, recreational behaviors, and their cultural contexts at specific locations (Calcagni et al., 2022; Goldspiel et al., 2023; Li et al., 2023; García-Mayor et al., 2025; Huang et al., 2025). However, regarding spatial representation, social media data still face two major challenges: (1) Lack of Precise Georeferencing: Textual data often lack explicit geolocation information, making it difficult to accurately characterize the spatial distribution of CES. (2) Constraint by Platform Spatial Units: The structure of the data is constrained by the platform's inherent spatial unit divisions. When the study area does not align with the platform's predefined boundaries, data acquisition becomes significantly restricted, compromising the accuracy and applicability of the research.

To overcome the limitations described above, this study integrates social media data with the SoLVES model to construct an analytical framework that combines user-generated subjective expressions with spatial modeling capabilities. This approach enables the customization of the study area independent of social media platform partitioning standards, thereby overcoming spatial boundary constraints. Simultaneously, it facilitates the seasonal dynamic assessment of CES, enhancing the temporal and spatial resolution of existing research. Consequently, this study aims to address the following two core research questions: (1) How do spatial preferences for CES change across different seasons? (2) Which environmental variables significantly influence public perception and evaluation of CES value? This research not only expands the spatiotemporal representation of CES perception but also provides a methodological innovation for the dynamic assessment of ecosystem services based on big data. Furthermore, it offers valuable practical insights for informing climate-adaptive planning and the development of seasonal management strategies for urban blue-green spaces.

2. Methods

2.1. Study site

Shanghai is a global first-tier city located in eastern China at the mouth of the Yangtze River, characterized by a subtropical monsoon climate. This study focuses on the North Section of Shanghai Botanical Garden (NS-SBG) as the research area. The botanical garden is situated in the southern part of Shanghai's central urban district. Established in 1974, Shanghai Botanical Garden covers an area of 81.86 ha (ha) and functions as a comprehensive botanical garden integrating plant introduction and acclimatization, horticultural display, scientific research, and public education. The NS-SBG is a newly renovated and expanded area of the botanical garden, developed between 2020 and 2023. It was officially opened to the public on September 28, 2023, with a total area of approximately 14.3 ha. Positioned as the "Shanghai Grand Garden", this section emphasizes the display of rare and endangered plants, native plants, and specialty ornamental plants. It features the city's first exhibition cold greenhouse and includes eight specialized gardens, such as: Bulbs and Perennials Garden; Iris Garden; Mediterranean Garden; Rock Garden (among others). These specialized gardens exhibit significant differences in spatial structure, landscape composition, and plant phenophases (seasonal appearances), providing an ideal sample base for analyzing seasonal variations in CES within this study (Fig. 1).

2.2. Research workflow

Fig. 2 illustrates the data collection and analysis workflow employed in this study. The research involved manually interpreting photo content to identify their geographic location of capture. Social media images confirmed to have been taken within the study area were then selected. Subsequently, public perception information related to CES was extracted from these images. Multiple sources of environmental data were collected, processed, and interpreted to derive the required environmental layers for the analysis. Finally, the environmental data and the perception data were jointly imported into the SoLVES model. This integration, combined with Kernel Density Estimation (KDE), was used to explore the characteristics of seasonal variations in CES within the blue-green space.

2.3. Data preparation

Existing research suggests that social media photo analysis is an excellent potential method for assessing CES (Tian et al., 2021). This study selected user-posted comment images from the social media platform Dianping as the primary data source. As the platform does not

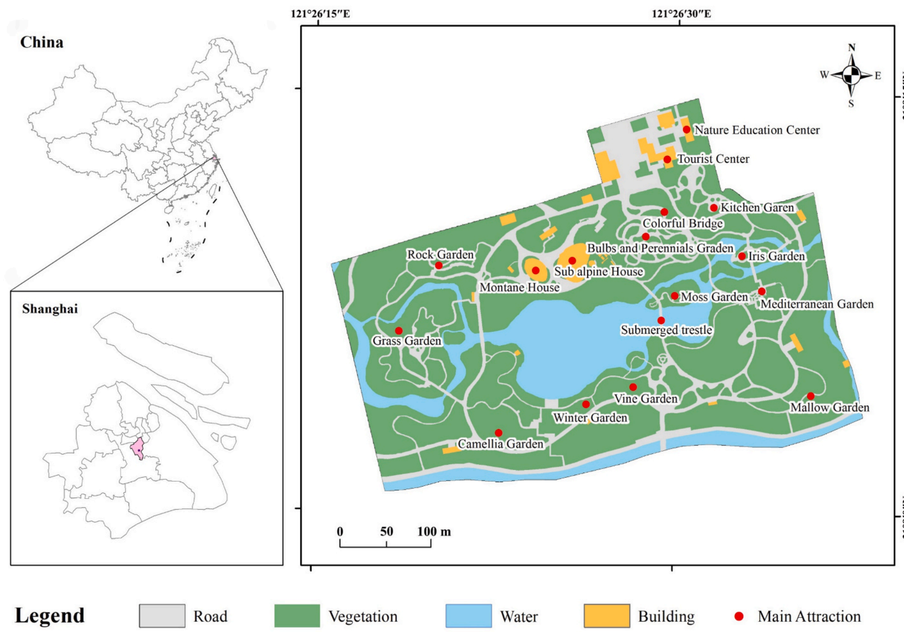

Fig. 1. Study area and main attractions.

differentiate between the North Section and other areas under the "Shanghai Botanical Garden" tag, the research team manually screened the photos. This involved interpreting photo content and background and comparing them against high-resolution aerial imagery captured by an unmanned aerial vehicle (UAV) to identify valid images taken specifically within the North Section. The data covered the period from October 2023 to September 2024, selected to encompass a full annual cycle and capture seasonal variations (spring, summer, autumn, and winter) in visitor activities. Images deemed irrelevant to the study, duplicates, or blurry were excluded. For multiple similar photos taken by the same user at the same location, only one representative photo was retained. A total of 1,288 valid photos were obtained. The photo locations were manually geolocated based on their background features and landscape elements, and the latitude and longitude coordinates of the shooting sites were extracted. Based on a synthesis of the literature and practical identification experience (Table 1), the CES type(s) represented in each photo were identified and classified. A single photo could correspond to multiple CES types simultaneously. The CES types were finally recorded in a table along with the timestamp of the photo's associated comment. (The research team consisted of three members: two students and one supervisor, all with professional backgrounds in landscape architecture or ecology. The tasks of photo geolocation and CES identification were carried out independently by the two students. Cases that were difficult for an individual to judge were resolved through team discussion to reach a consensus-based decision.) Considering Shanghai's location within a subtropical monsoon climate zone, characterized by distinct seasons with short spring and autumn seasons and long summer and winter seasons, this study divided the time period into four seasons: Autumn (October to November), Winter (December to February), Spring (March to May), and Summer (June to September). Photos were assigned to seasons based on this classification to ensure temporal comparability in the subsequent CES analysis.

For environmental data, we utilized three complementary sources: AutoNavi remote sensing map (obtained online), Official visitor guide map provided by the botanical garden website, High-resolution aerial imagery captured by UAV during field surveys. These three datasets were used in concert for reference to enable the extraction of more precise environmental data layers for the study area in subsequent steps.

2.4. *Data conversion and extraction*

The analysis in this study was conducted using SoLVES model version 4.1. The requisite spatial data were processed and prepared using ArcMap 10.4.1 and QGIS 3.8.2. The SoLVES model requires loading geospatial data and attribute table data into a PostgreSQL database. The spatial data used include vector data (Shapefiles) and raster data (Rasters) (Table 2).

Vector data consist of CES mapping points derived from social media photos and study area boundary data. Based on tabular photo data obtained from social media, each CES type within every photo was digitized into a CES mapping point using the photo's latitude and longitude coordinates, generating 3,125 CES mapping points for the entire year. These points underwent seasonal classification and were processed in a GIS platform to produce annual and seasonal CES mapping point datasets (SURVEY_POINTS). Study area boundary data (STUDY_AREA) were digitized by comparing the North Section's location on visitor guide maps with remote sensing imagery.

Four environmental variables were selected: Distance to Roads (DTR), Distance to Water Bodies (DTW), Distance to Main Attractions (DTMA), and Land Use/Land Cover (LULC) (Huang et al., 2023; Tian et al., 2024). Through manual interpretation integrating AutoNavi remote sensing maps, UAV high-resolution imagery, and visitor guides, layers representing paved roads, vegetation, water bodies, and main attractions were extracted. Euclidean distance calculations were performed to generate corresponding environmental raster datasets.

For each comment, the Score for each CES type was calculated using the formula:

$$SCORE_{(singleCEStype)} = (n/N)*100$$

where n = occurrence count of that CES type in the comment's photos, and N = total occurrences of all CES types in those photos. All CES scores were compiled into the CES Score Table (VALUE_ALLOCATION), which served as input for the SoLVES model.

2.5. *Investigating seasonal differences in CES perception in urban blue-green spaces*

2.5.1. *Kernel Density Estimation (KDE)*

Kernel Density Estimation (KDE) is a statistical method commonly

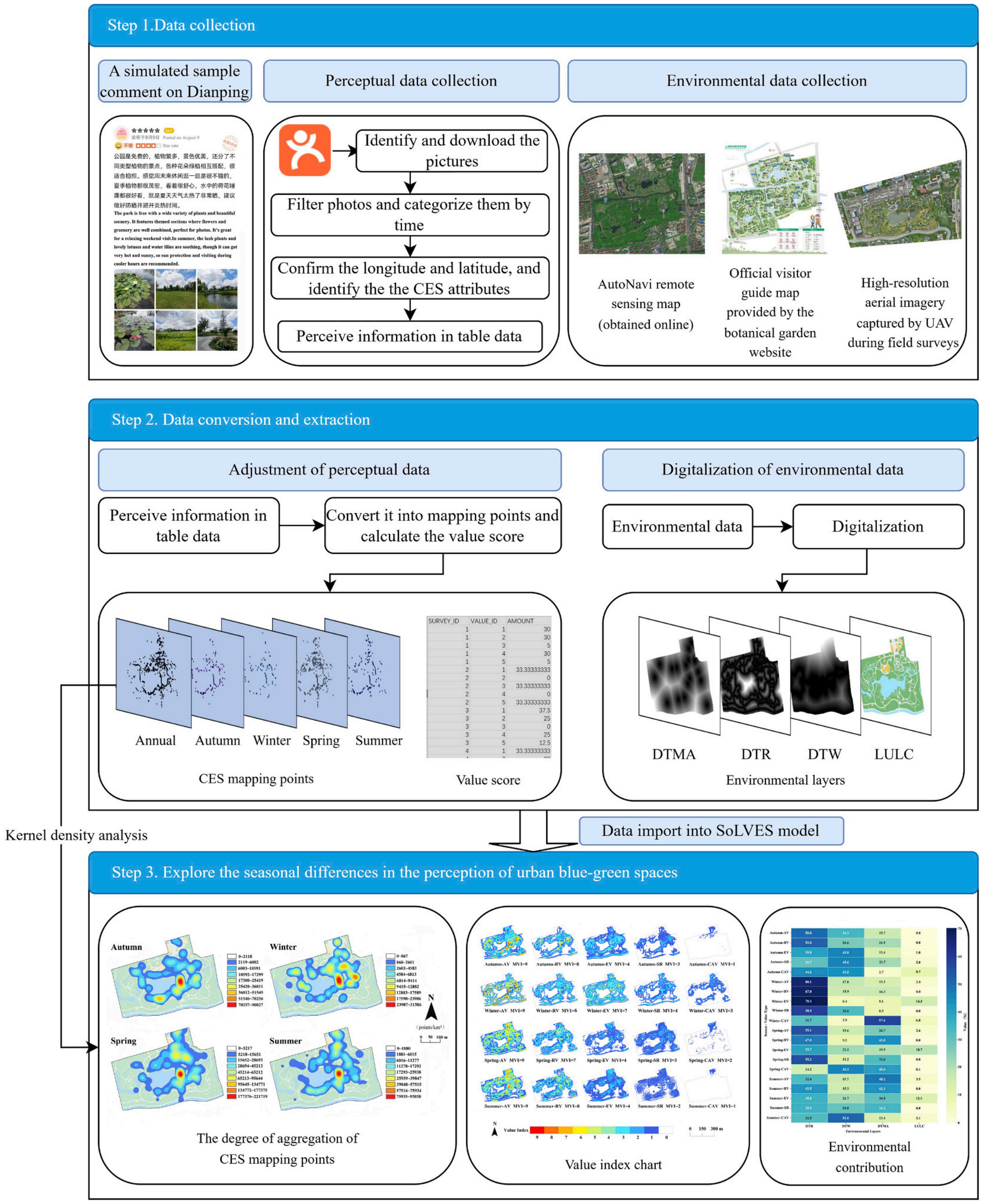

Fig. 2. Research workflow.

used for spatial point pattern analysis, capable of revealing spatial aggregation characteristics. This study employed KDE to perform visual analysis on the annual and seasonal CES mapping points, aiming to visualize differences in the spatial distribution of CES across seasons. To accommodate the relatively small spatial scale of the study area, the kernel search radius was set to 30 m. This parameter value was chosen to balance spatial resolution with visualization effectiveness.

2.5.2. SoLVES modeling and value quantification

This study employed the SoLVES tool developed by the USGS to model and quantify CES throughout the entire year and for individual seasons. SoLVES integrates with the Maxent (Maximum Entropy) modeling software, combining statistical modeling and GIS technology to quantify the relationships between CES values and environmental variables (Sherrouse and Semmens, 2020). The SoLVES analytical workflow proceeds as follows: (1) Conduct a weighted kernel density analysis on the mapped points for each CES type using their respective CES value scores, generating kernel density surfaces. (2) Standardize the kernel density surface of each CES to generate a 10-point, kernel-density based Value Index (VI) integer grid, and record its Maximum Value Index (MVI). (3) Apply Maxent to generate logistic value surfaces for each type of CES using mapped point data and environmental data. (4) Multiply the Maxent logistic output by its corresponding MVI and round the result to the nearest integer to generate a final integer raster (range 0–10), depicting the spatial distribution of the CES Value Index. (5) Extract model evaluation metrics, including the Area Under the Curve (AUC) value, Percent contribution, Permutation importance, and Jackknife test results. These metrics assess model performance and identify the influence mechanisms of environmental variables on CES distribution. An AUC value exceeding 0.7 indicates good model predictive capability (Sherrouse and Semmens, 2020).

Table 1
Explanations and photo identification criteria for CES value types used in this study.

CES Type	Definition	Photo Identification Features	References
Aesthetic Value (AV)	Pleasurable visual experiences provided by urban blue-green space landscapes.	Structures, boardwalks, plazas, floral bridges, islets, riverbanks, grassy slopes, flower beds, woodlands and similar landscape features.	Tian et al., (2021); Guan et al., (2023); Huang et al., (2024)
Recreational Value (RV)	Leisure activities (static/dynamic) in urban blue-green spaces.	Playgrounds, walking, picnicking and similar recreational activities.	Guan et al., (2023); Huang et al., (2023); Tian et al., (2024)
Educational Value (EV)	Acquisition of scientific knowledge in urban blue-green spaces.	Signage, interactive exhibits, nature museums and similar educational facilities.	Tian et al., (2021); Guo et al., (2023); Cao et al., (2025)
Social Relations (SR)	Opportunities for social interaction meeting public needs.	Groups engaged in social activities.	Gould and Lincoln (2017); Oteros-Rozas et al., (2018); Li et al., (2020)
Cultural& Artistic Value (CAV)	Landscapes awakening cultural awareness or inspiring artistic creation.	Cultural museums, heritage walls, artistic installations and similar cultural artifacts.	Gould and Lincoln (2017); Li et al., (2024); Yang et al., (2025)

Table 2
Environmental layers required for the SoLVES model.

Environmental Layer	Format	Description and Data Source	Accuracy
SURVEY_POINTS	Shapefiles	Public-perceived CES mapping points obtained from photographs	–
STUDY AREA	Shapefiles	Custom-defined study area delineated using tourist guide maps and remote sensing imagery	–
DTR	Raster	Distance to roads	1 m
DTW	Raster	Distance to water bodies	1 m
DTMA	Raster	Distance to main attractions	1 m
LULC	Raster	Land use/land cover	1 m

3. Results

3.1. Overall spatial perception characteristics of CES in the urban blue-green space

3.1.1. Spatial distribution characteristics of mapping points across the full year

Fig. 3 presents the results of the KDE for the annual CES mapping points within the NS-SBG. Overall, the distribution exhibits a distinct single-core agglomeration pattern. CES perception points are highly concentrated around the core scenic attraction, the “Submerged trestle”, forming a band-like feature encircling the central water body. Areas distant from the water body show a marked decrease in density. Furthermore, a spatial disparity exists between the eastern and western parts of the garden. Attractions are densely distributed in the eastern section, resulting in a significantly higher spatial density of CES mapping points compared to the relatively open and sparser western area.

3.1.2. Spatial distribution of the annual VI of CES

Fig. 4 presents the spatial distribution maps of the VI for the five categories of CES across the full year, simulated by the SoLVES model. Ranked by their MVI, the importance of the five CES categories is as follows: AV (9) > RV (6) > EV (4) > SR (2) > CAV (1). The results indicate that AV and RV are the most prominent in annual perception. The VI for AV exhibits the broadest spatial distribution, covering most areas of the garden section, with blank areas appearing only over the central lake surface, within off-limits lawn zones, and in peripheral regions. High VI is predominantly distributed along pathways and shows distinct clustering near scenic attractions.

Although the VI for RV, EV, and SR is relatively low, their spatial distribution trends largely coincide with that of AV, indicating a certain commonality in the spatial perception across CES types. In the “Northern Park Entrance” area, relatively high VIs were recorded for both AV and EV, while the VI for RV was significantly lower. This suggests that although this area offers rich visual landscapes and strong cognitive value, it is underutilized for recreation. CAV has the smallest distribution extent, forming only a localized cluster near the northern park entrance, with limited overall perceived value.

3.2. Seasonal distribution characteristics of CES in urban blue-green spaces

3.2.1. Seasonal variation in public perception activity

Fig. 5 and Table 3 present statistics on social media comment counts

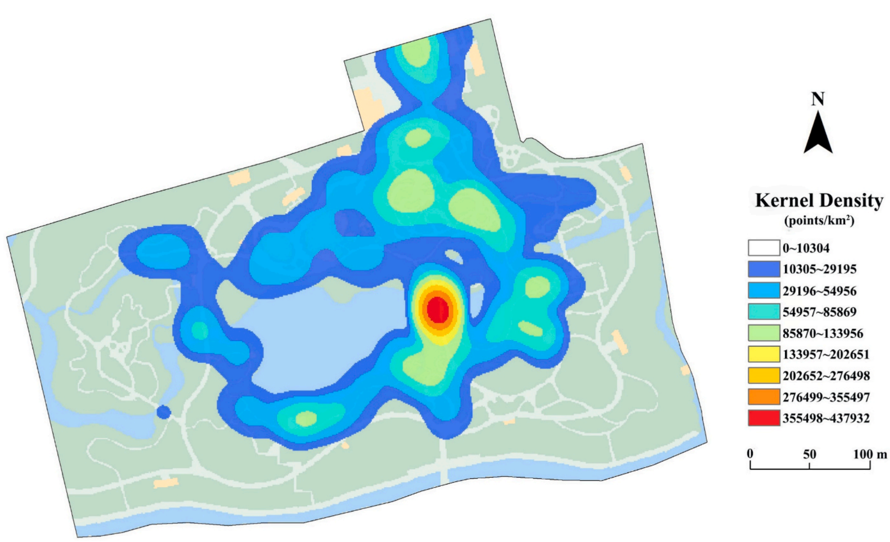

Fig. 3. KDE of CES mapping points across the full year.

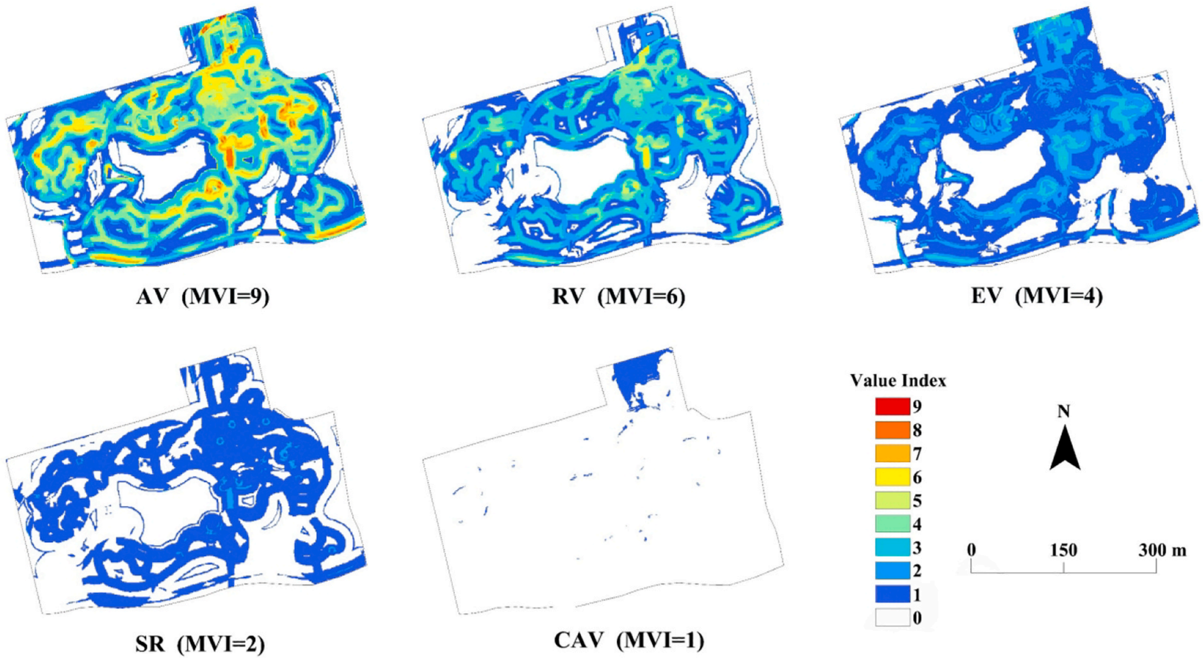

Fig. 4. Spatial distribution of the full-year VI of CES. *Note.* MVI (Maximum Value Index).

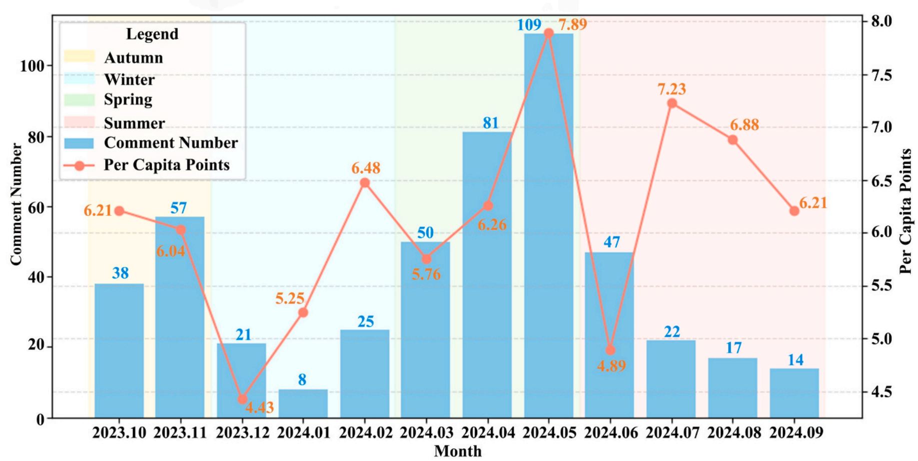

Fig. 5. Monthly comments and average number of mapping points in the study area.

Table 3
Full-year and seasonal counts of comments and mapping points in the study area.

Stage	Valid Social Media Comments	Monthly Average Valid Comments	Number of Mapping Points	Mapping Points per Capita
Full Year	489	40.75	3125	6.39
Autumn	95	47.5	580	6.11
Winter	54	18	297	5.5
Spring	240	80	1655	6.9
Summer	100	25	593	5.93

and mapping point characteristics from October 2023 to September 2024. The results reveal that: Spring exhibited the highest public perception activity, with a monthly average of 80 comments and 6.9 mapping points per capita. Winter showed the lowest activity, with only 18 monthly comments on average and 5.5 mapping points per capita. Autumn and Summer showed no substantial difference in per capita mapping points, though Autumn had significantly higher monthly comment counts. Monthly extremes further highlight climatic influences: May recorded the maximum comments (109), January had the minimum (8). This pattern demonstrates significant impacts of climate and seasonal variations on public CES perception behavior.

3.2.2. *Seasonal spatial distribution of mapping points*

Fig. 6 illustrates the KDE distribution of CES mapping points across different seasons. The Submerged trestle consistently served as a high-frequency perception hotspot throughout all four seasons, demonstrating its stable spatial attractiveness. During autumn and winter, mapping points were relatively dispersed, with perception clusters occurring on both eastern and western banks of the central water body. In contrast, during spring and summer, distribution significantly concentrated toward the eastern part of the garden, where perception

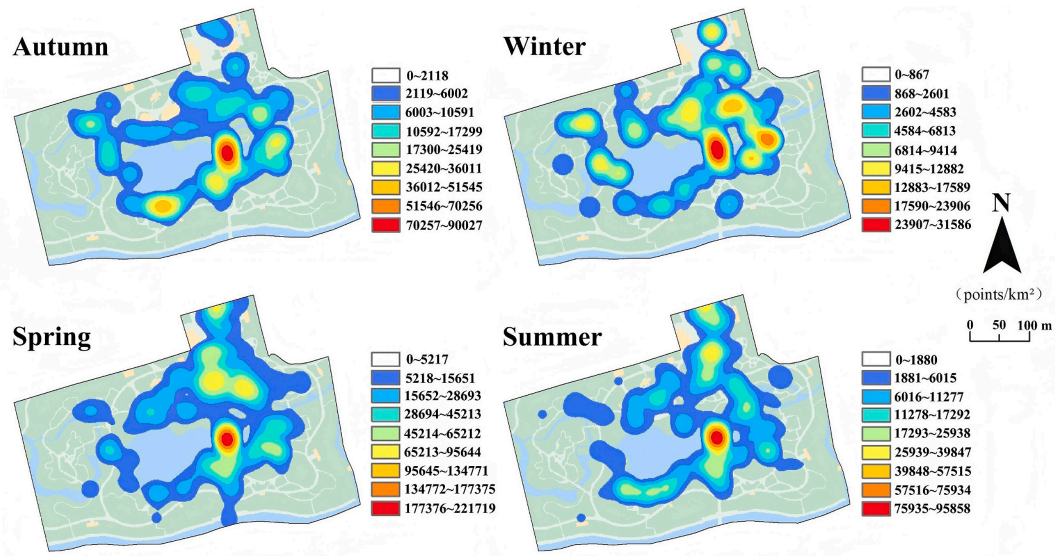

Fig. 6. KDE of CES mapping points across four seasons.

density along the eastern bank of the central water body far exceeded that of the western bank. This indicates that public activities became more spatially concentrated, with perception behaviors oriented toward specific scenic attractions.

3.2.3. Spatial distribution of the seasonal VI of CES

Table 4 lists the MVI for each of the five CES categories across seasons. With the exception of winter, the seasonal ranking of CES perception remained consistent: AV > RV > EV > SR > CAV. During winter, however, EV (MVI = 7) exceeded RV (MVI = 6), indicating distinct seasonal perception preferences.

Fig. 7 displays the spatial VI distribution maps for each CES category across the four seasons. Regardless of season or the full year, areas surrounding The Submerged trestle, Iris Garden, and the southern exit section of the Colorful Bridge consistently exhibited high-value clusters, demonstrating their cross-seasonal attraction. In spring and summer, VI significantly increased in the northern garden section (e.g., Bulbs and Perennials Garden), indicating that plant-display landscapes provide greater aesthetic and recreational value during warm seasons. In winter, high VI became highly concentrated along pathways, as degradation of herbaceous vegetation substantially reduced perception value in areas distant from roads. During summer, the distribution of VI expanded around water-adjacent areas (e.g., nearshore zones around the central lake), reflecting an enhanced public preference for aquatic landscapes during hot weather. It should be noted that due to AUC values below 0.7 during modeling, the spatial distribution results for EV in autumn and SR in summer are excluded from further analysis in this study.

3.3. Influence of environmental variables on CES spatial distribution

Figs. 8 and 9 present the Permutation Importance indices of environmental factors derived from the Maxent model for both the full year and individual seasons. Annual analysis results reveal that, excluding CAV, the primary influencing factors for all CES categories follow this order: DTR > DTMA > DTW > LULC. This demonstrates that DTR and DTMA are the dominant influencing factors in annual assessments, with their contribution rates generally exceeding 30 %.

Based on seasonal analysis, DTR consistently serves as a significant influencing variable for all CES categories except CAV, with contribution rates generally exceeding 30 %, particularly during winter. DTMA exhibits enhanced explanatory power for CES in spring and summer, with its contribution to AV surpassing 48 % in summer. DTW plays a relatively greater role for certain CES types during autumn. LULC demonstrates overall low contributions across CES categories, but provides meaningful explanatory power (contribution > 10 %) for EV in specific seasons. For CAV, characterized by its singular perception zone (primarily concentrated near the northern park entrance), no clear pattern of environmental contributions was observed.

These findings demonstrate that the driving mechanisms of environmental factors on CES spatial distribution exhibit distinct seasonal variations. The guiding effects of spatial elements including roads, attractions, and water bodies on public perception behaviors undergo dynamic changes across different periods.

Table 4
Seasonal MVI for CES.

CES Type	Autumn MVI	Winter MVI	Spring MVI	Summer MVI
AV	9	9	9	9
RV	8	6	7	8
EV	4	7	4	4
SR	3	4	3	2
CAV	1	3	2	1

4. Discussion

4.1. Perceived characteristics of CES and their influencing factors

The spatial distribution of CES is co-driven by natural and social factors, exhibiting significant spatial heterogeneity. This study demonstrates that while spatial perception varies across different CES types, it also reveals notable synergies—consistent with previous research findings (Du et al., 2024; Simeon and Wana, 2025). Huang et al. (2024) observed that "aesthetic and recreational values do not occur in isolation and that tourists tend to view them in combination." In this study, AV, RV, and EV exhibit strong spatial overlap. This suggests that visually compelling landscapes simultaneously enhance leisure experiences and stimulate learning interest, thereby amplifying educational engagement. Furthermore, educational activities may reciprocally sharpen aesthetic perception, creating a positive feedback loop among multiple CES values. Our observation of this phenomenon strongly supports the view of Wood et al. (2024) that considerable overlap can exist between different CES categories.

Despite relatively high scores in AV and EV at the North Section Entrance, RV remains notably low, indicating that public activities concentrate primarily within the garden's interior. Urban development

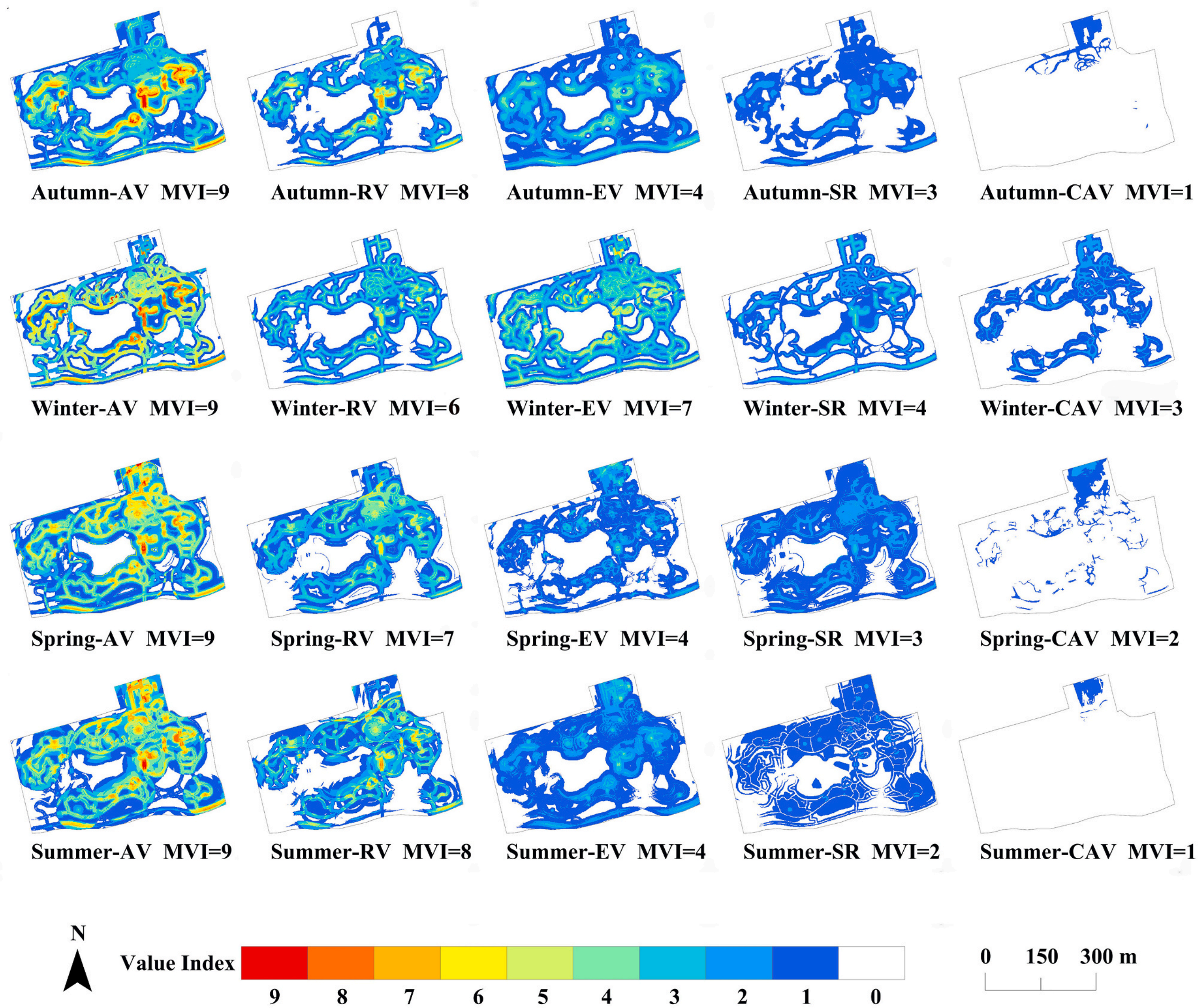

Fig. 7. Spatial distribution of the VI of CES across four seasons. *Note.* MVI (Maximum Value Index).

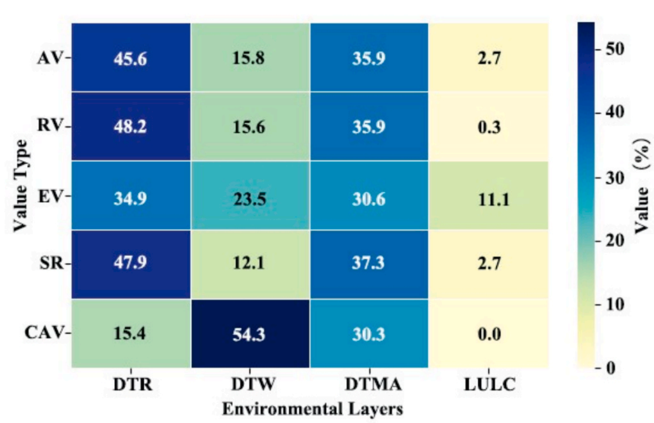

Fig. 8. Permutation Importance of environmental layers in the full-year analysis.

of social relations relies on deliberate physical space design and planning (Salih and Ismail, 2017), an aspect potentially underdeveloped in this area. Consequently, the VI for SR is generally low, showing only a marginal increase in zones of high aesthetic value. CAV typically requires dedicated cultural infrastructure. Due to limited such resources in this area, observable CAV clusters exist exclusively near the engraved entrance stones and visitor center at the northern entrance, with no significant aggregation detected elsewhere.

Recent research underscores the significant role of wetlands in providing Cultural Ecosystem Services (Wood et al., 2024; Jupe et al., 2025b). In our study, the distinct clustering of CES mapping points around the central water body corroborates the “water attraction hypothesis” (Li et al., 2020), demonstrating that aquatic landscapes significantly shape the spatial expression of CES by attracting public attention.

Field surveys at the “Submerged Trestle” and its vicinity attribute its consistently high CES perception values across all seasons to high landscape diversity, abundant water resources, rational plant arrangement, and complete facilities. Furthermore, its superior accessibility—stemming from a central location that serves as a pivotal link connecting the northern and southern sections of the garden—is a critical contributing factor. This case-specific finding highlights that accessibility is an important factor influencing CES perception (Lu et al., 2019; Benati et al., 2024). Yao et al. (2025) emphasized that improving urban green space accessibility enhances residents’ experiences of recreational, aesthetic, and mental health services. VI distribution maps reveal CES absence in “inaccessible zones” such as floral bed areas,

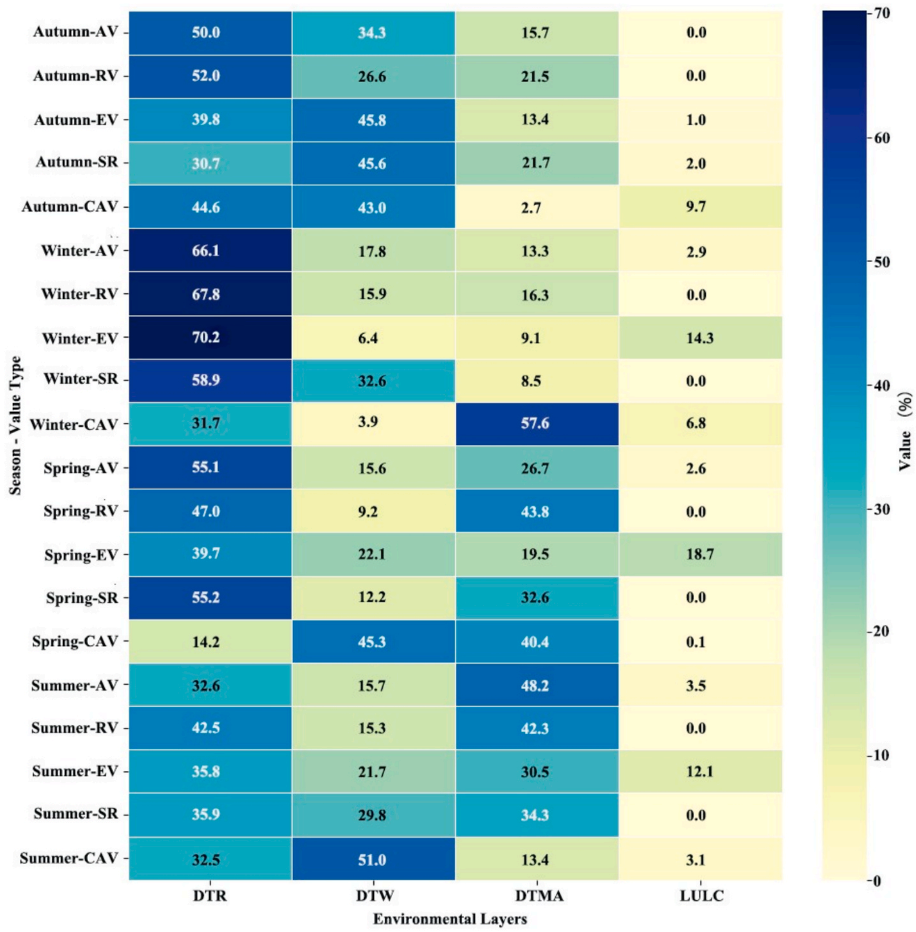

Fig. 9. Permutation Importance of environmental layers in seasonal analyses.

central lake surfaces, and peripheral edges, aligning with restricted visitation routes. Environmental variable analysis further confirms Distance to Roads (DTR) as the primary influencing factor for most CES types, with its dominance particularly pronounced in winter, validating accessibility's essential role in CES perception.

4.2. Seasonal regulation of CES perception

Public perception of CES exhibits significant seasonal variations across temporal dimensions. Consistent with existing research (Li et al., 2025a; Zhao et al., 2025), this study observed peak social media comments and mapping points during spring, reflecting richer CES experiences in this season. This phenomenon is driven by both the pleasant climate and floral abundance, and the catalytic effect of major events such as the "Shanghai International Flower Show". It should be noted, however, that the spatial distributions of AV and RV show striking similarity between spring and the full-year analysis, likely attributable to the dominant representation of spring data in annual aggregations, thereby skewing the annual results.

Summer CES perception demonstrated a distinct "hydrophilic clustering" pattern, characterized by an expanded spatial distribution of VI around water-adjacent areas. This suggests a public preference for waterscapes and aquatic plant landscapes, which may also be linked to the need for microclimate regulation during hot weather (Gao et al., 2023). Winter exhibits overall diminished perception intensity and a "dispersed perception" configuration, where VI distribution concentrates primarily along garden pathways and main routes, indicating climatic constraints on public mobility in cold conditions.

Plant phenophases significantly influence CES perception (Graves et al., 2017; Duan and Li, 2022). Within the Bulbs and Perennials Garden, spring and summer feature high landscape richness and lush vegetation, with many plants in bloom enhancing their appeal and significantly elevating perception values (Junge et al., 2015). Conversely, autumn and winter witness plant dormancy and diminished scenic value, correspondingly reducing perception intensity. Furthermore, environmental variable analysis reveals that although Distance to Main Attractions (DTMA) exerts a strong influence on CES, this effect is predominantly observed in spring and summer, while it becomes less pronounced in autumn and winter. Given that most attractions in the botanical garden are plant-based, this pattern further substantiates the regulatory role of plant phenological changes on CES perception. Notably, EV stands out as relatively high in winter compared to other seasons, potentially benefiting from the consistent exhibition content of the greenhouse. As an enclosed space less susceptible to seasonal fluctuations, the greenhouse provides critical support for winter CES perception. Furthermore, the lower CES perception values near the entrance and visitor center in autumn may relate to incomplete facility development during the initial opening phase, suggesting that study outcomes are sensitive to data temporal context.

4.3. Implications and recommendations for planners

The identified seasonal variations in CES exhibit distinct spatial specificity and management implications, supporting landscape optimization and refined management of blue-green spaces. First, this study confirms that pathway accessibility critically influences public CES perception. Green space design should enhance connectivity between path systems and major scenic attractions while improving access to peripheral zones and core landscapes, thereby mitigating cultural service gaps caused by poor accessibility (Carlier and Moran, 2019; Zhang et al., 2019). Second, CES values demonstrate pronounced seasonal fluctuations, with peak intensity in spring but diminished winter perception. Management strategies should implement "seasonally differentiated" landscape configurations based on plant phenology, including enhanced crowd management and traffic control during spring visitation peaks, strengthened water safety measures (e.g., warning signs, barriers) in summer, and organized seasonal festivals combined with indoor educational exhibits during autumn/winter to sustain visitor engagement. Third, core landscapes like the Submerged Trestle, integrating visual aesthetics, accessibility, aquatic interaction, and supporting facilities, maintain consistently high CES performance year-round. Future designs should emulate such multi-integrated high-CES concentration zones to elevate holistic experiential value. Finally, limited CAV and SR stem from inadequate spatial carriers; managers should introduce public art installations, temporary exhibitions, and interactive facilities, while planning dedicated social spaces (e.g., open lawns, circular seating arrangements) to enhance social interaction and cultural ambiance (Li et al.,2025b).

4.4. Research potential and limitations

This study integrated social media imagery with the SoLVES model, overcoming the limitations of social platform geographic partitioning through manual geolocation, enabling precise identification of CES in the northern section of the botanical garden. The method demonstrates good transferability and generalizability, applicable to blue-green space perception research across multiple regions and platforms. However, the following limitations remain: First, data scale and timeliness are constrained; due to the relatively short operational period of the northern section, social media sample size was limited, resulting in lower AUC values for some models. Moreover, using only one year of data means seasonal comparison results may be influenced by random fluctuations; future research should incorporate multi-year data to enhance robustness. Second, the granularity of the analysis requires improvement. The current scoring method struggles to differentiate between distinct value preferences within a single photograph; for instance, distinguishing aesthetic from educational content in images of greenhouses remains ambiguous. Furthermore, the level of detail in the Land Use/Land Cover (LULC) classification was constrained by manual interpretation, which consequently led to its diminished contribution within the model. Subsequent research could explore the integration of AI image recognition and deep learning techniques to enhance the precision and accuracy of the analysis (Yang and Zhang, 2024; Xu et al., 2025). Third, sample representativeness exhibits bias. The demographics of social media users may not match those of the wider society, with some groups being over- or under-represented (Sadah et al., 2015). Moreover, relying solely on photographs to represent CES may inherently bias the data towards aesthetic value (Huang et al., 2024). It is recommended that future studies combine text data, questionnaires, and interviews to strengthen the inclusivity and breadth of the research.

5. Conclusion

This study innovatively integrated the SoLVES model with social media image data, achieving precise delineation of the northern section of Shanghai Botanical Garden and spatiotemporal analysis of CES perception through geolocation methods, thereby addressing the limitation of ambiguous regional identification in traditional social media research.

The results reveal significant seasonal variations in public perception of CES in blue-green spaces. During summer, areas near water bodies exhibit prominent advantages; in winter, CES values markedly decline in deep vegetation zones distant from roads. Influenced by plant flowering seasons, CES values are substantially higher at sites with seasonal blooms during spring and summer. Roads and accessibility are critical factors affecting CES perception; core landscape areas such as the "Submerged trestle" consistently demonstrate high CES values across all seasons due to superior landscape design. Furthermore, annual data analysis tends to be overshadowed by dominant seasons, underscoring the importance of incorporating seasonal segmentation.

Based on the findings, this study recommends that planners optimize blue-green space design and management strategies according to seasonal characteristics: enhancing visitor flow guidance and transportation support in spring, improving waterfront safety facilities in summer, and introducing seasonal landscapes/activities to boost off-season appeal in autumn and winter. Concurrently, integrating cultural-artistic elements and social space facilities can strengthen public perception of CAV and SR.

This research provides empirical support and practical pathways for the seasonal management of urban blue-green spaces and the optimization of CESs, demonstrating notable theoretical value and application potential.

CRediT authorship contribution statement

Jie Li: Writing – original draft, Visualization, Software, Conceptualization, Writing – review & editing. **Zhenfeng Yang:** Writing – original draft, Software, Methodology, Data curation, Writing – review & editing. **Jiankang Guo:** Software, Resources, Data curation, Writing – review & editing. **Jia Zhou:** Resources, Data curation, Writing – review & editing. **Jie Wang:** Resources, Data curation, Writing – review & editing. **Kun He:** Writing – original draft, Supervision, Funding acquisition, Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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